

SULINA, Romania

WMO: 153600

Lat: 45.1624N

Lon: 29.7266E

Elev: 44

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	20.1	10.9	9.6	17.0	15.0	11.8	21.7	45.3	32.1	39.9	31.4	22.8	0	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	84.0	75.2	82.1	74.0	80.5	73.1	77.9	81.6	76.0	80.1	74.5	78.9	8.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	139.5	80.5	74.7	130.0	79.0	73.0	122.6	77.8	41.3	81.5	39.5	80.0	38.1	78.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.5	31.6	27.2	DB	10.5	88.6	5.6	2.2	6.5	90.2	3.2	91.5	0.1	92.7	-4.0	94.3	(4)
(5)			WB	9.9	79.2	5.5	2.2	6.0	80.8	2.7	82.1	-0.4	83.4	-4.4	85.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	33.6	35.4	41.7	51.3	62.3	70.4	74.7	74.4	66.2	56.3	47.1	37.8		
	DBStd	15.79	7.54	7.10	5.86	5.15	5.15	4.63	3.89	4.25	5.31	6.33	7.22	7.52		
	HDD50	1762	509	409	263	43	1	0	0	0	0	19	136	382		
	HDD65	4772	974	829	723	410	113	10	1	1	50	278	538	845		
	CDD50	3350	0	0	4	83	383	612	765	756	485	213	47	2		
	CDD65	887	0	0	0	0	31	172	301	292	85	6	0	0		
	CDH74	4014	0	0	0	0	21	523	1658	1698	113	1	0	0		
	CDH80	465	0	0	0	0	1	42	188	231	3	0	0	0		
(14) Wind	WSAvg	13.6	14.8	15.1	15.7	13.8	12.5	12.3	11.3	11.7	13.0	13.9	14.4	15.4		
(15) Precipitation	PrecAvg	10.9	0.7	0.7	0.6	0.7	1.0	1.2	0.9	1.1	1.1	0.7	0.9	0.9		
	PrecMax	19.2	2.9	3.3	2.5	2.8	5.6	4.7	3.5	5.1	5.0	4.5	3.7	4.5		
	PrecMin	4.3	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0		
	PrecStd	2.8	0.6	0.6	0.5	0.5	1.0	0.9	0.8	1.1	1.1	0.7	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	51.5	59.1	67.3	76.5	83.9	85.9	87.3	79.4	72.2	62.6	55.6		
		MCWB	45.4	46.0	49.7	59.8	67.4	72.7	75.1	76.3	70.1	67.5	58.9	52.5		
	2%	DB	46.7	47.3	53.1	63.1	73.2	80.9	83.7	84.5	76.7	69.2	60.3	52.5		
		MCWB	44.6	44.0	47.8	57.7	66.3	72.1	74.9	76.0	69.9	65.2	57.9	50.3		
	5%	DB	44.7	45.1	50.2	60.0	71.1	78.8	82.1	82.5	75.1	67.2	58.2	50.1		
		MCWB	42.9	43.0	46.3	55.3	65.2	71.1	74.3	74.3	68.9	63.4	55.7	48.0		
	10%	DB	42.8	43.6	48.3	57.6	69.2	77.0	80.6	80.8	73.5	65.0	56.3	47.9		
		MCWB	41.6	42.0	45.5	53.6	64.0	69.9	73.6	73.1	68.0	61.1	54.2	46.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	47.0	51.4	61.6	70.2	76.3	79.9	81.0	73.9	69.2	60.0	53.2		
		MCDB	48.0	49.7	56.2	65.8	74.0	80.3	82.7	83.9	76.0	71.1	61.5	54.7		
	2%	WB	44.9	44.8	48.7	58.4	67.8	74.3	77.9	78.6	72.3	66.2	58.1	50.8		
		MCDB	46.3	46.4	51.5	62.3	71.4	78.7	81.3	82.7	74.7	68.3	59.8	52.0		
	5%	WB	43.2	43.5	47.1	56.1	66.2	72.8	76.1	76.4	70.7	63.9	56.3	48.3		
		MCDB	44.4	44.8	49.3	59.1	69.8	77.3	80.1	80.5	73.5	66.6	57.7	49.7		
	10%	WB	41.7	42.1	46.0	54.4	64.8	71.3	74.7	74.6	69.2	61.6	54.7	46.4		
		MCDB	42.7	43.3	48.0	57.0	68.5	75.7	79.2	79.0	72.3	64.3	56.0	48.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	7.2	6.9	6.3	6.9	8.1	8.8	8.9	8.2	7.6	7.2	8.0		
		MCDBR	8.8	9.6	10.6	9.4	9.5	10.5	10.1	10.5	9.2	8.7	7.6	9.1		
	5% WB	MCWBR	6.9	6.6	6.6	5.9	6.0	6.3	6.0	6.5	6.2	6.7	5.4	7.2		
		MCWBR	8.0	8.0	9.0	7.8	8.2	8.7	8.5	8.6	8.0	7.9	6.3	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.380	0.408	0.437	0.416	0.414	0.432	0.439	0.413	0.391	0.368	0.351		
	taud		2.319	2.253	2.209	2.166	2.315	2.351	2.311	2.280	2.317	2.340	2.335	2.329		
	Ebn at Noon		233	248	259	261	271	272	265	258	254	242	225	220		
	Edh at Noon		26	33	39	45	40	38	40	39	35	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		369	614	1057	1555	1978	2115	2100	1869	1346	825	435	317		
	RadStd		62	87	106	135	125	116	103	99	108	74	65	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+110
(47) Regional (15 neighbors)		+1.03	N/A	N/A	+1.30	N/A	N/A	N/A	-238	+200	+139

Nomenclature: See separate page